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MIL-V-53112/1(ME)
10 January 1992

MILITARY SPECIFICATION SHEET

VEHICLE MAGNETIC SIGNATURE DUPLICATOR (VEMASID) SYSTEM,

M109/M992 FAMILY OF VEHICLES,

AN/VSQ-3(4)

This specification is approved for use within the US Army Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use within the distribution limitations noted below.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-V-53112.

APPLICABLE DOCUMENTS:

Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5865

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Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Box, Shipping, Fiberboard.
PPP-C-1842	-	Cushioning Material, Plastic, Open Cell.
PPP-T-60	-	Tape: Packaging, Waterproof.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-B-117	-	Bags, Sleeves and Tubing - Interior Packaging.
MIL-V-53112	-	Vehicle Magnetic Signature Duplicator (VEMASID) System, AN/VSQ-3, General Specification for.
MIL-A-53113	-	Amplifier, Vehicle Magnetic Signature Duplicator (VEMASID) System, AN/VSQ-3.
MIL-A-53114	-	Armor, Vehicle Magnetic Signature Duplicator (VEMASID) AN/VSQ-3(4).
MIL-C-53115	-	Coil, Vehicle Magnetic Signature Duplicator (VEMASID) AN/VSQ-3, General Specification for.
MIL-C-53115/1	-	Coil, Vehicle Magnetic Signature Duplicator (VEMASID) System, M109/M992 Family of Vehicles, AN/VSQ-3(4).
MIL-C-53116	-	Control Panel, Vehicle Magnetic Signature Duplicator (VEMASID) System, AN/VSQ-3.

STANDARDS

MILITARY

MS20392	-	Pin, Straight, Headed, Drilled Shank.
MS24665	-	Pin, Cotter (Split).
MS27183	-	Washer, Flat - Round, Steel.
MS51959	-	Screw, Machine Flat Countersunk Head.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWINGS

ME

TA13228E5014	-	VEMASID: AN/VSQ-3(V), M109A6.
TA13228E4987	-	Coil Assembly, M109.
D13228E4760	-	Control Panel Assembly.
D13228E4841	-	Armor Segment, M109 Forward and Rear.
D13228E4842	-	Armor Segment, M109 Side.
D13228E4911	-	Amplifier Assembly.
TA13228E4974	-	Cable Assembly, Special Purpose, Electrical, Branched: Control Panel, HCI.
TA13228E5033	-	Cable Assembly, Power, Electrical, HCI.
TA13228E5034	-	Cable Assembly, Power, Electrical, Branched, HCI.
TA13228E5037	-	Cable Assembly, Special Purpose, Electrical: Coil, Straight, HCI.
D13229E3347	-	Bracket, Sliding, Control Panel.
D13229E3359	-	Structural Section, Special Shaped: Hinge Support.
D13229E3361	-	Link Assembly, Hinge.
D13229E3435	-	Pivot Block, Lower.
D13229E3498	-	Bracket, Mounting, Control Panel Assembly.
D13229E3612	-	Cushioning Material Set, Packaging.
D13229E3626	-	Box, Shipping, VEMASID: M109.
D13229E5145	-	Amplifier Assembly Test Values.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-NDF, Fort Belvoir, VA 22060-5606).

Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for drawings), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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REQUIREMENTS

Component characteristics. The components of the type I system shall comply with MIL-V-53112, MIL-A-53113, MIL-A-53114, MIL-C-53115, MIL-C-53115/1 and MIL-C-53116.

Description. The type I system shall be as shown on top assembly TA13228E5014, and as specified in MIL-V-53112 and herein.

QUALITY ASSURANCE PROVISIONS

Functional performance. During system tests, when verification of system operation is required, the following procedure and criteria shall be used. If the system is turned off, turn the system on and verify that the two red fault lamps turn on for approximately two seconds and then turn off and remain off. Then proceed to verify the audible alarm function and output magnetic field as specified below. If the system is already turned on, verify that neither of the two red fault lamps are lit. Verify correct lamp function by pressing each red lamp housing and verifying that the lamp lights only as long as its housing is depressed. Next, verify audible alarm operation in accordance with MIL-C-53116. Then verify that the system output current has the frequency specified on drawing D13229E5145 for vehicle I.D. code 0110 (AAV). For the system reliability test only, using an oscilloscope, also verify that the system output current peak amplitude is between 28 and 51 amperes.

Electrical connection. Electrically connect the components of the system to each other and connect the system to power for the electromagnetic interference, reliability, and nuclear electromagnetic pulse tests in accordance with MIL-V-53112, figure 1. Use cable assemblies TA13228E4974, TA13228E5033-5, TA13228E5034-1, and TA13228E5037-2 and circuit breaker M13516/7-4, with the following exception: TA13228E5037-2 must be modified as close as practical to the coil end of the cable so that the required current amplitude and frequency measurements can be made. These measurements may be accomplished using a current probe or a precision shunt connected to the oscilloscope in accordance with MIL-V-53112, figure 1. If a precision shunt is used, its resistance shall not exceed 25 milliohms.

Inspection of packaging. Packaging inspection shall be in accordance with MIL-V-53112 and as further specified herein.

Examinations. Packaging of the system shall be examined for the following defects.

101. Cushioning material not as specified.
102. Tape not as specified.
103. Waterproof bag not as specified.
104. Wrap incomplete or not taped securely.
105. Absence or mis-orientation of assembled parts.
106. Missing item or wrong quantity of items.
107. Container not as specified.
108. Packs missing or not as specified.

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109. Blocking and bracing not as specified.
110. Improper closure of container.
111. Strapping not as specified.
112. Marking illegible, incorrect, or incomplete.

PACKAGING

Preservation. Preservation shall be level A as specified.

Level A.

1. The amplifier assembly (see D13228E4911) shall be preserved in accordance with method IIc of MIL-P-116. The bag shall conform to MIL-B-117, type I, class E, style 1. Cushioning material shall be in accordance with PPP-C-1842, type I or II.

2. Attach the control panel sliding bracket (see D13229E3347) to the bottom of control panel assembly (see D13228E4760) with four flat head screws conforming to MS51959-30. The relief groove on the bracket fits over the control panel housing flange. Tighten screws securely.

3. Attach the control panel assembly mounting bracket (see D13229E3498) onto the top of the control panel assembly with four flat head screws conforming to MS51959-30. Tighten screws securely.

4. The control panel assembly shall be preserved in accordance with MIL-P-116, method IIc. The cushioning material shall conform to PPP-C-1842, type I or II, and the bag shall conform to MIL-B-117, type I, class E, style 1.

5. Attach one lower pivot block (see D13229E3435) to each link assembly hinge (see D13229E3361) using one straight headed pin conforming to MS20392-10C127 and split cotter pin conforming to MS24665-363, placing flat washer conforming to MS27183-24 onto the straight pin prior to inserting into the pivot block, and prior to inserting the cotter pin.

6. Remove the cotter pins, headed straight pins, and washers from the ends of the two link assembly hinges. With the hardware removed, attach the two link assemblies to the special shaped structural section hinge support (see D13229E3359) using the two headed straight pins and split cotter pins, placing one flat washer onto the pin prior to inserting it into the link assembly, two before inserting the pin into the structural channel, one after passing through the structural channel, and one prior to inserting the cotter pin.

7. The remaining miscellaneous hardware items, excluding the coil, armor, and hinge assembly, shall be preserved in accordance with method IA-8 of MIL-P-116. The cushioning material shall conform to PPP-C-1842, type I or II, and the bag shall conform to MIL-B-117, type I, class E, style 1.

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8. Small hardware items such as screws, washers, nuts, and pins shall be preserved in the same waterproof bag.

9. Place the bottom section of the packaging cushioning material set (see D13229E3612-1) with flat section down into a weather resistant corrugated box conforming to PPP-B-636, class WR, type CF, variety DW, grade 15C, 20x16x12.75.

10. Place the amplifier assembly into the corrugated box on top of the cushioning, fitting the handle into the appropriate cutout.

11. Place the top section of the cushioning material set into the corrugated box on top of the amplifier assembly with the flat section down.

12. Place the control panel assembly into one of the cutouts in the top section of the cushioning material set.

13. Dunnage with additional cushioning material where necessary to ensure minimal movement during shipment.

14. Close the corrugated box and seal with waterproof tape conforming to PPP-T-60, type III, class optional.

15. Consolidate the smaller, bagged items into a close-fitting corrugated box conforming to PPP-B-636, class weather-resistant. Select a size which will fit in the corners or between the coil supports at the back of the large shipping box (opposite side to front cushions, see D13229E3626-8). Dunnage with additional cushioning material where necessary to ensure minimal movement during shipping. Seal with waterproof tape conforming to PPP-T-60, type III, class optional.

Packing. Packing shall be level A or C as specified.

Level A.

1. Place the hinge support structural assembly into the shipping box conforming to PPP-B-601, overseas type (see D13229E3626) between the supports for the coil, at the front of the box (see figure 1).

2. Place the M109 coil (see TA13228E4987) into the shipping container, orienting the coil to fit properly on the coil supports with the down-turned coil edge resting on the front cushions (see D13229E3626-8). The electrical connector should be located by the support that is not positioned symmetrically with the others.

3. Place the M109 forward and rear armor segment (see D13228E4841) and M109 side armor segment (see D13228E4842) onto the coil in the appropriate locations.

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4. Place the corrugated box containing the amplifier assembly into the shipping box in the center section appropriately blocked for the box.

5. Orient cover to place the foam sections over the coil supports and place the cover on the shipping box.

6. Apply steel strapping conforming to ASTM D 3953, type 1, heavy duty, zinc-coated, 3/4-inches wide, and ASTM D 4675 in accordance with PPP-B-601 for a box with interior cleats.

Level C. The VEMASID system, AN/VSQ-3(4), and all of its components, preserved as specified, shall be packed as specified for level A, except the boxes shall be domestic class or type, whichever is applicable and the strapping shall not be zinc-coated.

Marking. In addition to any special marking specified in the contract or purchase order, marking shall be in accordance with MIL-STD-129.

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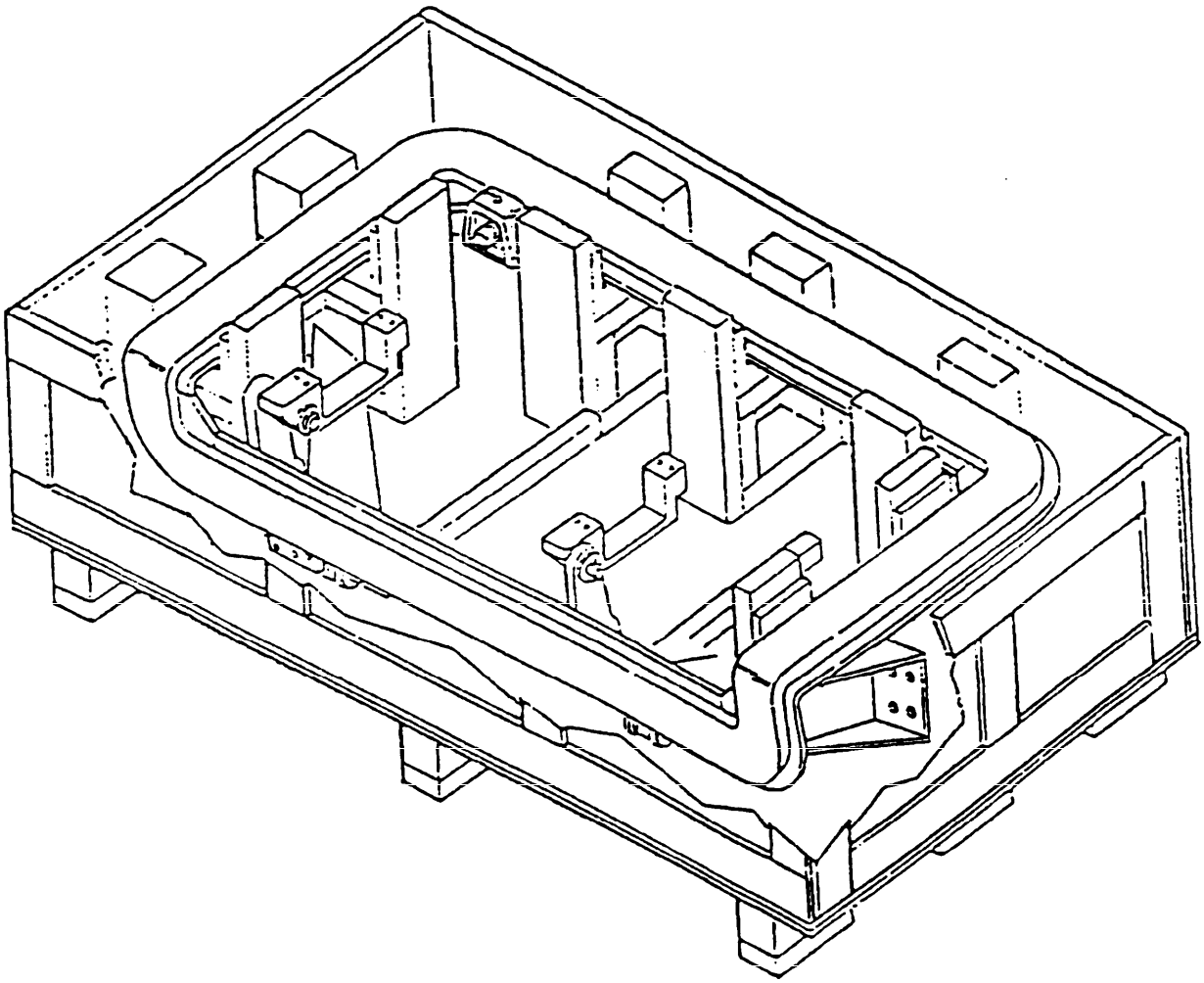


FIGURE 1. Packing of hinge support structural section.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-V-53112(ME)	92/01/10

3. DOCUMENT TITLE
Vehicle Magnetic Signature Duplicator (VEMASID) System, M109/M992 Family of Vehicles, AN/VSQ-3(4)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
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